

Sen Zhang

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Summary

Registered NY architect (NCARB), 9 years post-Master's, currently Computational BIM Developer at Ennead Architects. Solo developer behind Ennead's Design Technology stack: EnneadTab (450+ tools, three U.S. Copyright registrations, 60 daily users — roughly half the design staff), plus BimRunner, RenderPolisher, FieldDay, AIA-Sustainable-Reporting, and 6 other production services. Outside Ennead: Realm (real-estate networking platform) and several small-business ventures with AI agents handling daily operations. Speaker: Advancing Computational Design 2026, Autodesk University 2024, Digital Built Week Americas 2023.

Experience

Ennead Architects LLP / Computational BIM Developer

New York, NY · 2022–Present

Architectural Designer, BIM Specialist

2020–2022

- **Sole developer of every plugin and SaaS tool in Ennead's Design Technology stack (160-person firm), while also performing BIM manager duties on active projects. Platform development, BIM management, complex-geometry triage.**
 - + Developer of EnneadTab and 8+ other firm-internal tools — see Selected Works below
 - + BIM management on active projects:
 - Healthcare (NE US): NYU Langone, Northwell, Lenox Hill
 - Commercial (Asia): FTZ Shanghai, ByteDance Houhai HQ, Bilibili HQ
 - + Solution architect for large-model BIM data, complex families; in-house training and mentoring
 - + Rhino-to-Revit interoperability via zero-learning-curve internal tooling; triage of complex parametric and industrial-scale geometry

Arquitectonica / Designer, BIM Specialist, CG Visualization

New York, NY · 2017–2020

- 40 Hudson Yards (NYC office tower); Amazon HQ2 master planning (Chicago); 31 Parliament mid-rise residential tower (Toronto SmartCity)
- Curtis Block (Calgary); Warsaw Hotel (Poland); Newport Waterfront; Equinox Hotel (San Jose); Queens Sherbourne mixed-use with historical preservation (Toronto)

RPI Publication Dept. / Book-design, Asset Management

Troy, NY · 2013–2016

Speaking & Recognition

Advancing Computational Design Conference / Speaker

Austin, TX · 2026

The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency

Autodesk University / Session Speaker

San Diego, CA · 2024

Revit As A Game Engine

Digital Built Week Americas / Panel Speaker

Dallas, TX · 2023

Promoting Computational Design to Non-Programmers

Credentials & Technical Stack

- Registered Architect, New York (NCARB)
- Languages: Python (CPython, IronPython) — primary; TypeScript / React / Next.js — applied (LLM-assisted); Kotlin / Jetpack Compose
- BIM / 3D APIs: Revit API, RhinoCommon API, Grasshopper, Dynamo

- Web stack used in production: Next.js 14–16, NextAuth v5, Drizzle ORM, Vercel Postgres / Blob / KV, Resend, Chart.js, react-pdf, GitHub Actions, FastAPI
- AI integration: on-prem Ollama (private deployment, data stays inside firm network); cloud Claude / OpenAI / Gemini; Claude Code agents in production; model-agnostic via MCP
- Spoken: English, Mandarin Chinese (native)

Education

Columbia University GSAPP / M.S. Advanced Architecture Design	New York, NY · 2016–2017
Rensselaer Polytechnic Institute / Bachelor of Architecture	Troy, NY · 2012–2016
Tongji University / Major in Urban Planning	Shanghai · 2010–2012

Awards & Recognition

• EnneadTab — U.S. Copyright Office — 3 registrations, authorship credited to Sen Zhang	2024
• AEC Hackathon — Educational Tool Connecting Junior and Senior Students — Team — Honorable Mention (NYC)	2023
• Hyperloop Campus, Las Vegas — Honorable Mention	2020
• Greenway to Ideal City, Chengdu — 2nd Place	2019
• Tokyo Anti-Library — Honorable Mention	2018
• Non Architecture: Dance — Finalist	2018
• AIA Henry Adams Award	2016
• Magna Cum Laude Honors — Dean's List 2012–2016	2016
• George T. Droste Scholarship — Structural engineering study	2015
• Faculty Selection Award — Academic proficiency in Architecture	2015

SELECTED WORKS

Note on visuals — the EnneadTab product family and the internal Ennead-owned tools below are Ennead IP and cannot be included here with images. A live walkthrough with client and firm-internal details redacted is available during interview.

Office Tools at Ennead Architects

EnneadTab / Developer

2021–Present

Firm-wide plugin ecosystem across Revit, Rhino, CAD, Adobe, Excel, and Email. Built and maintained solo. Designed end-user-first: silent updates, no installer, no onboarding documentation. Reached firm-wide voluntary adoption without a top-down mandate or training rollout. 500+ tools, three U.S. Copyright registrations (2024), 60 active daily users, 17×/person/workday, 63,627 logged events Q4 2025. Stack: Python (CPython + IronPython) + Revit/Rhino APIs.

EnneadTab Self-Healing Agent Ecosystem / Developer

2025–Present

Autonomous self-healing layer wrapping the EnneadTab-OS codebase. Detects production errors across 60 daily users, drafts fixes, and opens pull requests ready for human review and merge. Featured in the Austin 2026 talk on agile tool introduction.

BimRunner / Developer

2026–Present

Production analytics platform for Revit project health. Local harvester ingests data from 700 models weekly; web dashboard surfaces historical scoring and trending data, project pin-up wall, scheduled cron reports, and an AI agent that generates custom analysis scripts on demand to answer complex cross-project queries. Used by project leads across the firm to triage live work before coordination crises.

RenderPolisher / Developer

2025–2026

AI-driven render post-processing pipeline with deeply customized features tailored to architects' language and delivery workflow. Cleans and uplifts architectural visualizations through a configurable model chain; integrates into the Ennead delivery workflow via the EnneadTab toolbar.

AIA-Sustainable-Reporting / Developer

2026–Present

CRON Excel pipeline + web dashboard for AIA 2030 sustainable-design metrics, with location-based filtering and project-to-project comparison. Replaces the spreadsheet that used to circulate quarterly for AIA 2030 reporting.

EnneadTabWiki / Developer

Ongoing

Public documentation portal for the EnneadTab tool family. Tool catalog, usage data, and reference content auto-update from the EnneadTab-OS source repository on every release — docs stay in lockstep with shipped tools, without manual rewrite. Includes an AI-assisted RAG system for semantic search across all tool docs and usage patterns.

EmployeeData / Developer

2025–2026

Decision-support system for Ennead partners on capital-equipment prioritization. Consolidated HR records (177 employees), weekly directory scrapes, and per-machine hardware reports into a single employee-by-machine view; surfaced that 70% of the 176-machine fleet was Retired and 81% of GPUs were 5+ years old, sequenced into a quarterly replacement plan through 2027 Q3. Replaced an ad-hoc Excel email workflow; gated to the partners and DT lead who approve capex. Delivered and retired April 2026 once the decision cycle completed.

EnneadTabLLM / Developer

2025–2026

On-prem AI assistant for plain-English queries across firm standards, templates, and project history. Client data never leaves the firm network. Basis of the 2026 Austin talk "Cloning the Expert."

FieldDay / Developer

2026–Present

Offline-capable web app for site inspection on phones, tablets, and desktops. PDF-based spatial pinning, photos, voice notes, and comments with zero-connectivity capture and cloud sync once back online. Preserves spatial accuracy across drawing revisions.

EnneadTabHome / Developer

2025–2026

Landing page and secure auth gateway for the EnneadTab application family at enneadtab.com — single sign-on via Microsoft 365 / Entra ID, fronting all in-house web services.

EnneaDuck / Designer

2022–Present

Community-mascot system across the EnneadTab ecosystem — pixel-art duck embedded throughout the toolset (achievements, status pings, gamified adoption hooks) plus a Next.js Duckland community page. Per user feedback, made the toolset feel more memorable and approachable — the warmth layer that helped a top-down rollout become a self-sustaining voluntary community.

2025 Performance Review / Developer

2025–2026

Firm-wide employee-evaluation system used by HR to streamline the annual review process. Excel ingest → interactive HTML report with software-proficiency tracking and visual analytics. Deployed to GitHub Pages.

Independent

Realm / Digital Strategy Architect

2025–Present

Networking platform connecting architects, real-estate operators, and exhibition partners through curated offline events. Multi-step RSVP, admin/client dashboards, role-based partner gating, branded email flows.

Atelier-T / Founder

2025–Present

Early-stage venture for high-end salons. Operating model pairs a customer-facing web app with an agent-team that runs daily small-business operations end-to-end.

Social-Media Ads Service / Lead Engineer

Ongoing

AI-heavy content creation paired with social-media engagement automation, packaged as an ad-service offering for live small-business clients.

Social-Media Channel — Android App / Lead Engineer

Ongoing

Native Android consumer app focused on intentional social-media consumption. Early-stage / pre-launch.

LakeHouse-Design / Developer

Ongoing

Collection of daily high-value workflow timesaver plugins for Rhino.